

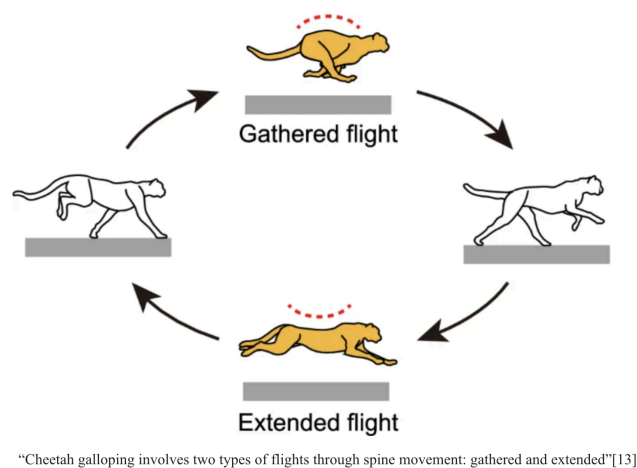
Reinforcement Learning Simulation of a Biomimetic Spined Quadruped Robot

Student: Rahul Rajesh

Faculty Member: Professor Jiefeng Sun

1. Introduction and Background

Legged animals demonstrate a remarkable ability to move efficiently across varied terrain, which is something engineers still struggle to replicate in robotics. Cats, in particular, achieve speed and agility through the motion of their flexible spine, which compresses and extends during each stride to generate power and balance. In contrast, most existing quadruped robots, such as MIT's Cheetah3 [2] or Unitree's A1, use rigid spines. This rigidity simplifies control but limits stride length, stability, and energy efficiency. Developing robots that better mimic biological spines could lead to machines that move faster, use less energy, and handle rough terrain more effectively.



To create controllers for such complex motion, this project uses reinforcement learning (RL). RL is a branch of machine learning where an artificial agent learns to make decisions by trial and error. Instead of relying on hand-coded control rules, RL algorithms learn directly from experience, improving performance over time by maximizing a "reward" that measures success [8]. RL has shown strong precedent for robot control problems, including walking, manipulation, and balancing [6], making it a natural approach for learning adaptive movements in spined quadrupeds.

Research in quadruped locomotion has progressed rapidly in the past decade. The MIT Cheetah 3 [2] and ANYmal [1] platforms demonstrated that combining physics-based simulation with machine learning enables fast and stable locomotion. However, these robots lack articulated spines, leaving a gap in understanding how spinal flexibility affects control. Biomechanics studies discuss the impact that a cat's spine can have on speed and propulsive energy [12], suggesting that similar mechanisms could enhance robotic motion efficiency.

Simulation tools such as MuJoCo (Multi-Joint dynamics with Contact) and Isaac Sim/Isaac Lab from NVIDIA provide realistic physics engines for studying these systems [7, 10]. MuJoCo offers high-precision rigid-body dynamics on CPUs, while Isaac Sim runs on GPUs and integrates reinforcement-learning libraries for large-scale, parallel training. Despite these advances, few studies directly compare RL performance for spined quadrupeds across these environments or explore how different spinal stiffness levels influence learned gaits [5].

2. Research Question

How can reinforcement learning be used to train a control policy for a spined quadruped robot? This study will test whether modern RL methods can learn stable, efficient locomotion for a robot whose body includes a flexible spine. Success would provide insight into how robotic

systems can leverage spinal motion to improve efficiency and robustness, which are capabilities that could influence future legged robots.

3. Research Plan and Methodology

This project will focus on simulation and algorithmic development rather than hardware fabrication. First, the Squad (spined quadruped) model will be created in both MuJoCo and Isaac Sim using CAD data from the Sun Robotics Lab. The model will include articulated joints representing the flexible spine, with physical properties such as mass distribution, gravity, and friction [11]. The simulation setup will be validated by checking that motion responses and contact forces behave realistically under simple scripted movements.

Next, reinforcement learning control will be implemented using NVIDIA IsaacLab, a learning framework built on Isaac Sim to accelerate robot policy training using GPU-accelerated simulations [7]. By using Isaac Lab's RL framework for Isaac Sim. Training will occur on ASU's SOL supercomputer, which allows multiple simulations to run in parallel.

A reward function will guide learning toward smooth, energy-efficient walking by rewarding forward velocity, symmetric gait, and low energy consumption while penalizing instability or excessive spinal bending. Training sessions will vary parameters such as spine stiffness and terrain roughness.

Performance will be evaluated through measurable metrics: learning speed, gait stability, total energy used per distance traveled, and torque efficiency at the joints. Comparative experiments will test (1) rigid-spine vs. flexible-spine models and (2) MuJoCo vs. Isaac Sim implementations. Learned gaits will be visualized and analyzed using Python tools. Depending on progress, results may later be transferred to the physical Squad platform for validation of real-to-sim consistency.

The outcome will be quantitative data showing how spinal flexibility influences locomotion learning and whether GPU-accelerated simulation offers advantages over traditional CPU-based methods. Ultimately, this research will yield design and control recommendations for future biomimetic robots capable of moving efficiently across complex terrain.

4. Alignment with Fulton Schools Research Themes

This project aligns with the Data research theme of the Ira A. Fulton Schools of Engineering. It advances data-centric engineering through the development and evaluation of reinforcement-learning algorithms and large-scale simulation data pipelines. Training and optimization require extensive processing of motion and performance datasets, directly reflecting Fulton's emphasis on computational tools, machine learning, and data-driven decision-making for engineering innovation.

Works Cited

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- [6] Li, Yuxi. “Deep Reinforcement Learning: A Brief Survey.”
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- [9] Tedrake, Russ R. *Underactuated Robotics: Algorithms for Walking, Running, Swimming, Flying, and Manipulation*. MIT OpenCourseWare, 2023, underactuated.csail.mit.edu
- [10] Todorov, Emanuel, et al. “MuJoCo: A Physics Engine for Model-Based Control.” *IEEE International Conference on Intelligent Robots and Systems (IROS)*, 2012
- [11] Wensing, P. M., et al. “Proprioceptive Actuator Design in the MIT Cheetah: Impact Mitigation and High-Bandwidth Physical Interaction for Dynamic Legged Robots.” *IEEE Transactions on Robotics*
- [12] Y.K Kim et al. “The Role of Relative Spinal Motion during Feline Galloping for Speed Performance”, *ScienceDirect*, [https://doi.org/10.1016/S1672-6529\(14\)60064-0](https://doi.org/10.1016/S1672-6529(14)60064-0)
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Personal Statement

As a rookie wildlife photographer, I have spent hours observing different animals leap and balance with precision that no machine has yet replicated. This fascination evolved into a scientific curiosity when I came back from India. I thought, “We see all these robots moving so stiffly. How can we capture natural movement in robotics?” My journey from photographing nature quickly changed to modeling natural movement computationally when I joined Sun Robotics Research Lab.

As a Computer Science and Mathematics (Statistics) major at Arizona State University, my coursework in probability, digital design, and object-oriented programming gives me a strong foundation for analytical thinking, while my research experiences turned that knowledge into action. My role at Sun Robotics Lab involves computation and mechanics, from implementing physics-based simulations in MuJoCo and Isaac Sim, to optimizing reinforcement learning (PPO) frameworks for gait efficiency and stability, ultimately teaching me that innovation in robotics often begins where biology meets computation.

My favorite part of my research is its potential to contribute to both engineering and biological understanding. By analyzing how spinal flexibility impacts locomotion efficiency, I hope to uncover design principles that can extend beyond robots. This project aligns closely with the Data research theme of the Fulton Schools, as it is the base of data-centered learning systems.

Beyond the lab, I bring curiosity and passion to my other endeavors as well. My experience as a coding instructor has helped me communicate complex ideas to students, reinforcing my belief that knowledge gains meaning when shared. Winning second place at the Devlabs Strategy Hackathon and working on sustainable projects like ReThread (sustainable fashion chrome extension) taught me the multitude of ways we can impact the world around us using tech. These experiences are my driving factor to not only solve technical problems but to pursue research that has real-world influence.

In my future, I plan to pursue graduate studies in robotics and artificial intelligence, envisioning a career at the intersection of research and design. Starting early with creating systems that move, adapt, and interact with the world as fluidly as the animals that inspire them, the FURI program represents an opportunity to deepen my research independence and contribute to a growing field that fuses biology, computation, and engineering into something fundamentally human.

16 Week Timeline:

	Month							
	Oct	Nov	Dec	Jan	Feb	Mar	April	May
Phase 1: Initial Familiarization and SetUp								
Task 1.1 – Continue background research on quadruped RL	■	■						
Task 1.2 – Review MuJoCo & Isaac Sim documentation; run test simulations on SOL	■	■						
Task 1.3 – Identify metrics for performance evaluation	■	■						
Phase 2 – Environment Implementation & Validation								
Task 2.1 – Update and finalize Squad URDF and physics properties		■	■					
Task 2.2 – Validate MuJoCo and Isaac Sim environments			■					
Phase 3 – RL Integration (Baseline)								
Task 3.1 – Implement PPO algorithm using Stable-Baselines3 on MuJoCo environment			■	■	■			
Task 3.2 – Design reward functions; test initial training runs for forward locomotion			■	■	■			
Phase 4 – RL Integration (Advanced)								
Task 4.1 – Integrate PPO within Isaac Lab for GPU-accelerated training on SOL					■	■		
Task 4.2 – Conduct baseline comparisons (learning rate, entropy, reward shaping)					■	■		
Phase 5 - Experimentation								
Task 5.1 – Run experiments: rigid vs. flexible spine; different stiffness settings						■	■	
Task 5.2 – Domain randomization tests (terrain friction, mass, joint noise)						■	■	
Phase 6 – Analysis and Evaluation								
Task 6.1 – Analyze reward curves, torque efficiency, gait patterns using Python						■	■	
Task 6.2 – Compare Isaac Sim vs. MuJoCo results; write preliminary findings							■	
Phase 7 - Reporting and Final Draft								
Task 7.1 – Draft FURI poster, writeup, and mentor review								■
Task 7.2 – Finalize and present results at FURI Symposium							■	■
Milestones:								
Literature Review and Complete Set Up		■						
Simulation Environments Validation			■					
Baseline PPO Controller Trained (MuJoCo)					■			
Port PPO Framework to Isaac Lab and Begin GPU-Based Training						■		
Conduct Comparative Experiments: Rigid vs. Flexible Spine							■	
Analysis and Poster Finalized								■

Budget Details:

FURI Budget Request		
Student Name:	Rahul Rajesh	
Faculty Mentor Name:	Jiefeng Sun	
Spring 2026		
Budget Request Item	Expense	Budget Justification
Compute credits & GPU Access	200	accelerate RL training and large scale simulation experiments
3d Printing Components	150	fabricate prototype spinal modules
Robot's obstacle course materials, misc.	50	setting up courses for robot to walk on and learn
Final Budget	\$400	

Arizona State University
Unofficial Transcript

Name: Rahul Rajesh
Student ID: 1232677526

Print Date: 05/13/2025
External Degrees
Desert Mountain HS
High School Diploma 05/23/2024

Test Credits					
2024 Sumr					
Course	Description	Attempted	Earned	Grade	Points
BIO 182	General Biology II		4.000	Y	
BIO 181	General Biology I		4.000	Y	
ASU	(Elective Credit)		3.000	Y	
CRDEXM					
MAT 270	Calculus w/Analytic Geometry I		4.000	Y	
STP 226	Elements of Statistics		3.000	Y	
Course Trans GPA:	0.000	Transfer Totals:	0.000	18.000	0.000

Test Credits					
2024 Sumr					
Course	Description	Attempted	Earned	Grade	Points
Course Trans GPA:	0.000	Transfer Totals:	0.000	0.000	0.000

Beginning of Undergraduate Record

2024 Fall					
Course	Description	Attempted	Earned	Grade	Points
ASU 101-CAI	The ASU Experience	1.000	1.000	A+	4.333
CSE 110	Principles of Programming	3.000	3.000	A+	12.999
FSE 100	Introduction to Engineering	2.000	2.000	A+	8.666
FSE 150	Grand Challenges for Engrg	3.000	3.000	A+	12.999
Note:	Honors Course				
HON 171	The Human Event	3.000	3.000	A	12.000
Note:	Honors Course				
MAT 266	Calculus for Engineers II	3.000	3.000	A	12.000
THF 160	Introduction to Storytelling	3.000	3.000	A+	12.999
		Attempted	Earned	Points	
Term GPA:	4.00	Term Totals	18.000	18.000	75.996
Cum GPA:	4.00	Cum Totals	18.000	36.000	75.996

Academic Standing: Good Standing
Term Honor: Dean's List

2025 Spring					
Course	Description	Attempted	Earned	Grade	Points
CSE 205	Object-Oriented Program & Data	3.000	3.000	A	12.000
EEE 120	Digital Design Fundamentals	3.000	3.000	A+	12.999
ENG 105	Adv First-Year Composition	3.000	3.000	A+	12.999
Note:	Honors Course				
HON 272	The Human Event	3.000	3.000	A+	12.999

Course	Description	Attempted	Earned	Grade	Points
Note:	Honors Course				
MAT 243	Discrete Math Structures	3.000	3.000	A+	12.999
MAT 267	Calculus for Engineers III	3.000	3.000	A	12.000
		Attempted	Earned	Points	
Term GPA:	4.00	Term Totals	18.000	18.000	75.996
Cum GPA:	4.00	Cum Totals	36.000	54.000	151.992

2025 Fall					
Course	Description	Attempted	Earned	Grade	Points
CSE 240	Intro to Programming Languages	3.000	0.000	NR	0.000
GPH 381	Geography of Natural Resources	3.000	0.000	NR	0.000
IEE 380	Prob & Stats Engrg Prob Solving	3.000	0.000	NR	0.000
MAT 300	Mathematical Structures	3.000	0.000	NR	0.000
STP 420	Introductory Applied Statistic	3.000	0.000	NR	0.000
STP 421	Probability	3.000	0.000	NR	0.000
		Attempted	Earned	Points	
Term GPA:	0.00	Term Totals	0.000	0.000	0.000
Cum GPA:	4.00	Cum Totals	36.000	54.000	151.992

END OF TRANSCRIPT

Rahul Rajesh

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EDUCATION

B.S. in Math (Statistics) , B.S. in Computer Science

Cumulative GPA: 4.0

Barrett, The Honors College, Arizona State University

Expected Graduation: May 2028

Honors/Awards: Runner Up @2025 Strategy x Devlabs Hackathon (50+ Teams), Dean's List, President's Scholarship

Relevant Coursework: Introductory Applied Statistics, Intro to Programming Languages, Object Oriented Programming, Multivariable/Vector Calculus, Discrete Math, Digital Design Fundamentals, Mathematical Structures, Probability

EXPERIENCE

Sun Robotics Research Lab (*Prof. Jiefeng Sun*)

Tempe, AZ

Researcher

Jul 2025 – Present

- Simulating dynamics, kinematics and control of a biomimetic spined quadruped robot using MuJoCo, Python, and C
- Leveraging ASU's supercomputer (SOL) to accelerate large-scale simulations in multiple terrain types and gaits to provide insights into the data and reinforcement learning frameworks as well as performance and adaptability

Mammal-Virus MetaData and Modeling

Tempe, AZ

Researcher (Prof. Beckett Sterner)

Aug 2025 - Present

- Curating and standardizing metadata from 1000+ peer-reviewed studies on rodent virus interactions to support computational modeling of zoonotic risk by extracting ~40 fields of ecological and experimental data per paper
- Upcoming work includes computational modeling and statistical prediction of host-virus networks using R/Python.

Solar Devils

Tempe, AZ

Integrations Team Member

Aug 2024 – May 2025

- Played a key role in data integration efforts to support a solar-powered electric vehicle to enhance performance
- Utilized SQLite3 & Python to create a database with storage and retrieval of data from the vehicle's CAN network to enable my team to make decisions regarding performance and troubleshooting while displaying on the vehicle's screen

iCode - Scottsdale

Scottsdale, AZ

Coding Instructor & Mentor

Aug 2023 – May 2024

- Facilitated coding sessions for students in areas such as drone programming, web design, and game development.
- Provide one-on-one mentoring to students to help overcome coding challenges and fostering problem-solving skills

PROJECTS

ReThread - AI-Powered Sustainable Shopping Extension (*2nd Place, DevHacks x Strategy*)

Tempe, AZ

Team Member

Mar 2025

- Developed a Chrome extension that integrates with major fashion retail sites to suggest sustainable alternatives from third party seller platforms (FB marketplace, Poshmark) in real time, to promote textile recycling and upcycling
- Utilized Google Cloud Vision for image recognition and Chrome Extension APIs for keyword product matching

Campus Compass - Campus Life Involvement Agent (*ScaleU + Principled Innovation Academy*)

Tempe, AZ

Team Member

Mar 2025

- Developed an AI agent chatbot using Python and AWS Bedrock to help students discover personalized campus activities and organizations based on their interests, major, and goals to support their adjustment to college life
- Scraped data from ASU's SunDevilSync platform to populate the chatbot's knowledge base hosted on an S3 bucket

Grand Challenges Scholars Program

Tempe, AZ

Security Theme Scholar

Jul 2024 – May 2025

- Led a team in designing a conceptual model for an Augmented Reality lightweight headset aimed at supporting Alzheimer's patients, with emphasis on active memory recall through uploaded photos/videos, safety and usability.
- Presented our project findings and design proposals to distinguished faculty and researchers at a convention, demonstrating the potential impact our idea can have on improving quality of life for individuals with Alzheimer's.

Autonomous MINDSTORM Robot Project

Tempe, AZ

Team Member

Aug 2024 - Dec 2024

- Designed and implemented an autonomous robot to simulate a real-world scenario of transporting a disabled passenger
- Programmed using MATLAB for autonomous navigation, obstacle avoidance, and precise pickup/drop-off capabilities.

SKILLS & INTERESTS

Technical Skills: Experience using Java, Python, C, HTML, SQLite3, MATLAB, LaTeX, Operating Systems (MacOS, Windows), VSCode

Other Skills: English, Tamil, Statistical Thinking for Data Science and Analytics (ColumbiaX Course), Problem Solving

Interests: Sports, Piano, Painting, Motorcycles, Dance, Hiking, Photography, Animal Shelter Volunteer (LostOurHome)